SIMON FRASER UNIVERSITY

CMPT 371

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Mini Project 2 - Results

TEAM 30

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Output

Server:

```
PS C:\Users\Miloni\CMPT371\MP2> & C:/Python312/python.exe
c:/Users/Miloni/CMPT371/MP2/server.py
Server listening on 142.58.211.109:12000
Received message from ('142.58.213.195', 61578): 0:Message 0
Received message from ('142.58.213.195', 61578): 1:Message 1
Received message from ('142.58.213.195', 61578): 2:Message 2
Dropping packet from ('142.58.213.195', 61578)
Received message from ('142.58.213.195', 61578): 4:Message 4
Received message from ('142.58.213.195', 61578): 5:Message 5
Received message from ('142.58.213.195', 61578): 6:Message 6
Received message from ('142.58.213.195', 61578): 3:Message 3
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 6:Message 6
Received message from ('142.58.213.195', 61578): 3:Message 3
Received message from ('142.58.213.195', 61578): 4:Message 4
Corrupted message from ('142.58.213.195', 61578), ignoring...
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 3:Message 3
Received message from ('142.58.213.195', 61578): 4:Message 4
Received message from ('142.58.213.195', 61578): 5:Message 5
Corrupted message from ('142.58.213.195', 61578), ignoring...
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 6:Message 6
Received message from ('142.58.213.195', 61578): 4:Message 4
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 5:Message 5
Received message from ('142.58.213.195', 61578): 6:Message 6
Received message from ('142.58.213.195', 61578): 5:Message 5
Received message from ('142.58.213.195', 61578): 6:Message 6
Received message from ('142.58.213.195', 61578): 7:Message 7
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 8:Message 8
Received message from ('142.58.213.195', 61578): 9:Message 9
Received message from ('142.58.213.195', 61578): 7:Message 7
```

```
Received message from ('142.58.213.195', 61578): 8:Message 8
Dropping packet from ('142.58.213.195', 61578)
Received message from ('142.58.213.195', 61578): 7:Message 7
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 9:Message 9
Received message from ('142.58.213.195', 61578): 7:Message 7
Received message from ('142.58.213.195', 61578): 8:Message 8
Received message from ('142.58.213.195', 61578): 9:Message 9
Received message from ('142.58.213.195', 61578): 7:Message 7
Corrupted message from ('142.58.213.195', 61578), ignoring...
Received message from ('142.58.213.195', 61578): 9:Message 9
```

Client:

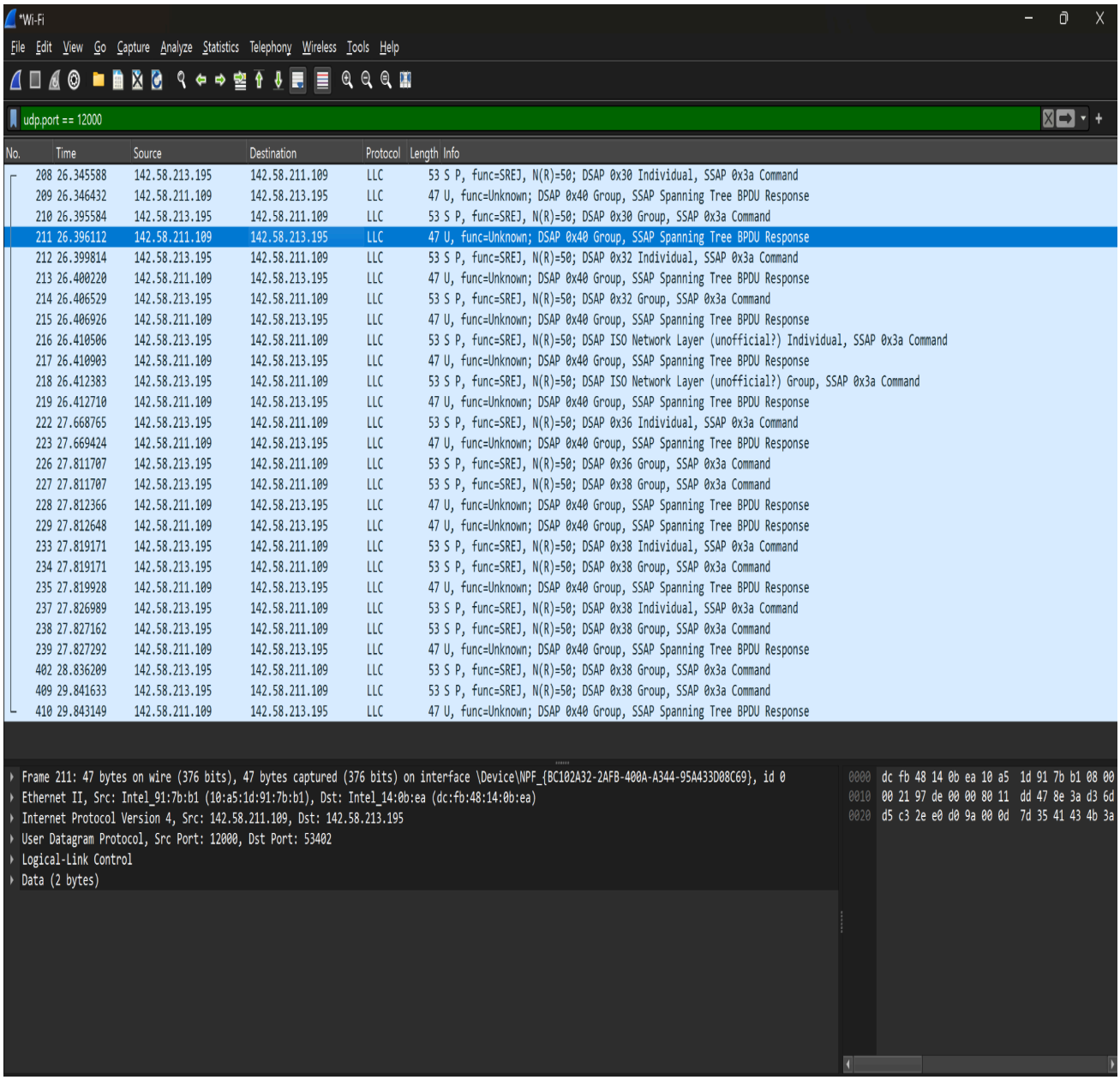
```
[Running] python -u "c:\Users\hitar\Downloads\MP2\client.py"
Sending message: b'0:Message 0'
Received ACK: ACK:0
Sending message: b'1:Message 1'
Sending message: b'2:Message 2'
Received ACK: ACK:1
Received ACK: ACK:2
Sending message: b'3:Message 3'
Sending message: b'4:Message 4'
Sending message: b'5:Message 5'
Sending message: b'6:Message 6'
Received ACK: ACK:4
Resending window: [3, 4, 5, 6]
Sending message: b'3:Message 3'
Dropping message: b'4:Message 4'
Sending message: b'5:Message\xdf5'
Sending message: b'6:Message 6'
Received ACK: ACK:5
Resending window: [3, 4, 5, 6]
Sending message: b'3:Message 3'
Sending message: b'4:Message 4'
Sending message: b'\xca:Message 5'
Sending message: b'\xc9:Message 6'
Received ACK: ACK:6
```

```
Resending window: [3, 4, 5, 6]
Sending message: b'3:Message 3'
Sending message: b'4:Message 4'
Sending message: b'5:Message 5'
Sending message: b'6:Message 6'
Received ACK: ACK:3
Received ACK: ACK:6
Resending window: [4, 5, 6]
Dropping message: b'4:Message 4'
Sending message: b'5:Mes\x8cage 5'
Sending message: b'6:Message 6'
Received ACK: ACK:3
Resending window: [4, 5, 6]
Sending message: b'4:Message 4'
Sending message: b'5:Message 5'
Dropping message: b'6:Message 6'
Received ACK: ACK:4
Received ACK: ACK:3
Resending window: [5, 6]
Sending message: b'5:Message 5'
Sending message: b'6:Message 6'
Received ACK: ACK:4
Resending window: [5, 6]
Sending message: b'5:Message 5'
Sending message: b'6:Message 6'
Received ACK: ACK:5
Received ACK: ACK:6
Sending message: b'7:Message 7'
Dropping message: b'8:Message 8'
Sending message: b'9:Message 9'
Received ACK: ACK:4
Resending window: [7, 8, 9]
Dropping message: b'7:Message 7'
Sending message: b'8:Message 8'
Sending message: b'9:Message 9'
Received ACK: ACK:5
Resending window: [7, 8, 9]
Sending message: b'7:Message 7'
Sending message: b'8:Message 8'
Sending message: b'9:Message 9'
```

```
Received ACK: ACK:6
Resending window: [7, 8, 9]
Sending message: b'7:Message 7'
Sending message: b'8:Mess\x9ege 8'
Sending message: b'9:Message 9'
Received ACK: ACK:5
Resending window: [7, 8, 9]
Sending message: b'7:Message 7'
Sending message: b'8:Message 8'
Sending message: b'9:Message 9'
Received ACK: ACK:6
Resending window: [7, 8, 9]
Sending message: b'7:Message 7'
Sending message: b'8:Mes\x8cage 8'
Sending message: b'9:Message 9'
Received ACK: ACK:7
Received ACK: ACK:8
Received ACK: ACK:9

[Done] exited with code=0 in 1.024 seconds
```

Wireshark Trace:



*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp.port == 12000

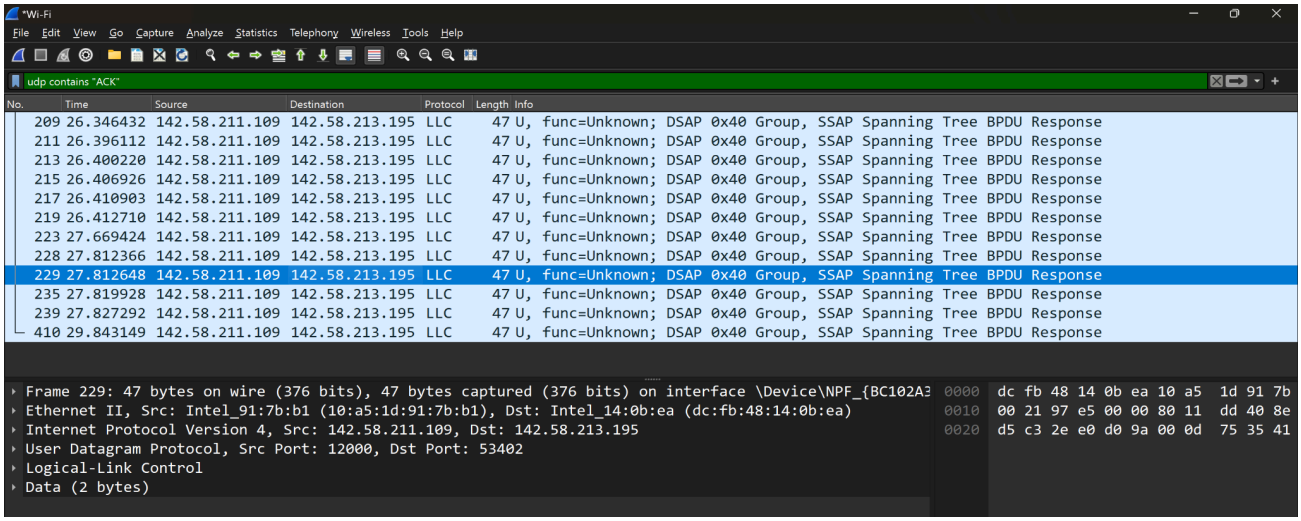
No.	Time	Source	Destination	Protocol	Length	Info
208	26.345588	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x30 Individual, SSAP 0x3a Command
209	26.346432	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
210	26.395584	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x30 Group, SSAP 0x3a Command
211	26.396112	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
212	26.399814	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x32 Individual, SSAP 0x3a Command
213	26.400220	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
214	26.406529	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x32 Group, SSAP 0x3a Command
215	26.406926	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
216	26.410506	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP ISO Network Layer (unofficial?) Individual, SSAP 0x3a Command
217	26.410903	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
218	26.412383	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP ISO Network Layer (unofficial?) Group, SSAP 0x3a Command
219	26.412710	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
222	27.668765	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x36 Individual, SSAP 0x3a Command
223	27.669424	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
226	27.811707	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x36 Group, SSAP 0x3a Command
227	27.811707	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Group, SSAP 0x3a Command
228	27.812366	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
229	27.812648	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
233	27.819171	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Individual, SSAP 0x3a Command
234	27.819171	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Group, SSAP 0x3a Command
235	27.819928	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
237	27.826989	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Individual, SSAP 0x3a Command
238	27.827162	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Group, SSAP 0x3a Command
239	27.827292	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response
402	28.836209	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Group, SSAP 0x3a Command
409	29.841633	142.58.213.195	142.58.211.109	LLC	53	S P, func=SREJ, N(R)=50; DSAP 0x38 Group, SSAP 0x3a Command
410	29.843149	142.58.211.109	142.58.213.195	LLC	47	U, func=Unknown; DSAP 0x40 Group, SSAP Spanning Tree BPDU Response

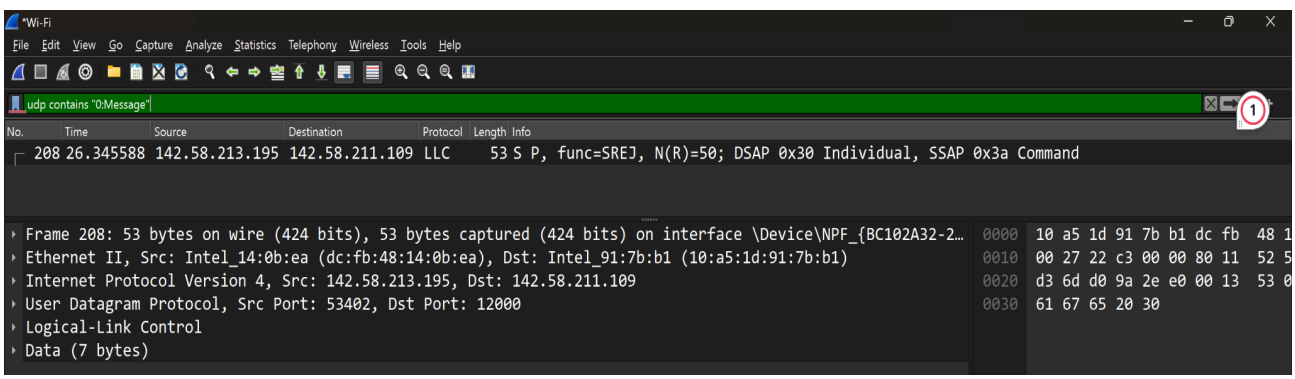
Frame 211: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface \Device\NPF_{BC102A32-2AFB-400A-A344-95A433D08C69}, id 0
Ethernet II, Src: Intel_91:7b:b1 (10:a5:1d:91:7b:b1), Dst: Intel_14:0b:ea (dc:fb:48:14:0b:ea)
Internet Protocol Version 4, Src: 142.58.211.109, Dst: 142.58.213.195
User Datagram Protocol, Src Port: 12000, Dst Port: 53402
Logical-Link Control
Data (2 bytes)

```
0000  dc fb 48 14 0b ea 10 a5  1d 91 7b b1 08 00
0010  00 21 97 de 00 00 80 11  dd 47 8e 3a d3 6d
0020  d5 c3 2e e0 d0 9a 00 0d  7d 35 41 43 4b 3a
```

ACKs

- Filtering out the packets which has acks



udp contains "0:Message"

- Filtering out the packets by 0 string as payload.